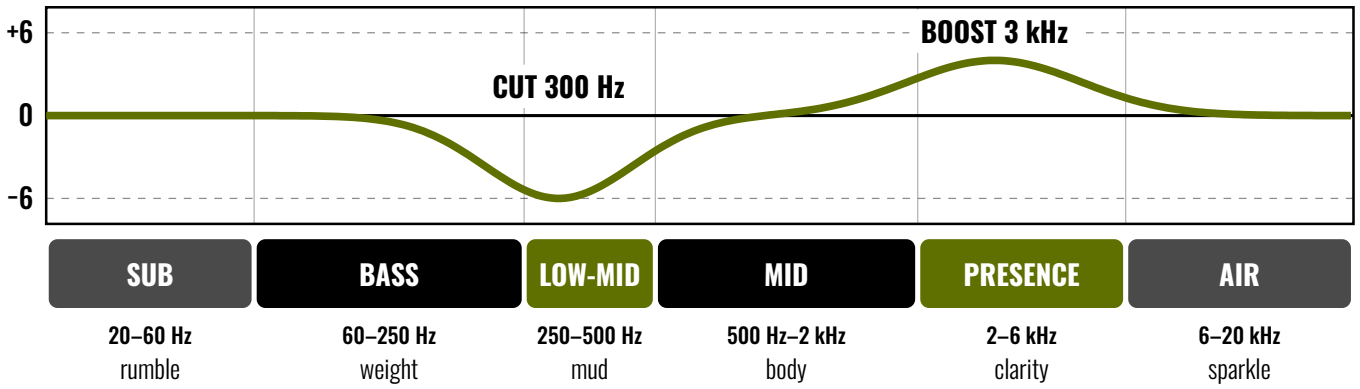
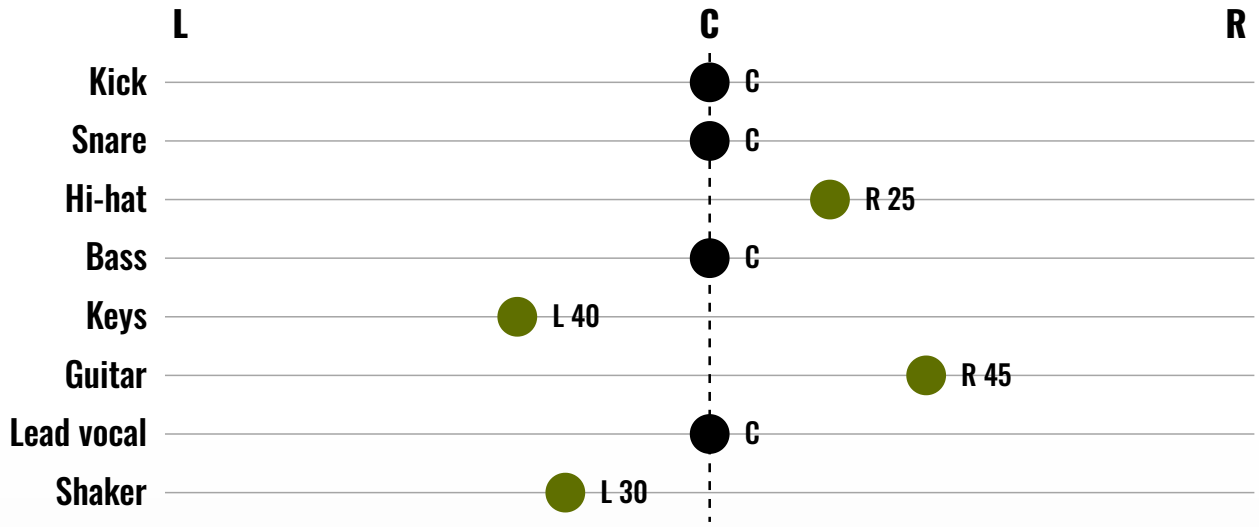
HUDSON BEAT





SHAPE TONE, DEPTH AND SPACE

FX DEEP DIVE

DMP2 · EQ · panning · reverb · delay · compression · intro slides

GRADES 9–12 · 7 CLASS PERIODS · 23 PAGES + 18-SLIDE DECK

FX DEEP DIVE

How to use this resource

START HERE

Teacher

WHAT THIS IS ·

A 7-class-period mixing project for high school music technology. Each student takes an existing track, either one of their own earlier projects or a multitrack from the Cambridge-MT library, and mixes it in Logic Pro with at least five different audio effects: EQ and panning on every track, plus effects such as compression, reverb, delay or modulation. It is a structured project from my Digital Music Production 2 course (focus: mixing and effects), and the projects in that course can be taught in any order.

QUICK START ·

1. Read the pacing guide (page 5) and my teacher notes (pages 6–7).
2. Decide where source tracks come from and pre-select sessions (page 4): no audio is included.
3. Print the student pages (pages 13–22), one set per student.
4. Open the intro deck and read the speaker notes for Class 1.

THE INTRO DECK ·

Every slide has a class: Class 1 (slides 1–6), Class 2 (slides 7–11), Class 3 (slide 12), Class 4 (slide 13), Class 6 (slides 14–16) and Class 7 (slides 17–18). Class 5 is a working day with no new slide. Every slide has speaker notes.

Fonts: the deck uses Oswald and Source Sans 3 (free on Google Fonts, built into Google Slides). If PowerPoint or Keynote shows a different font, install the two fonts or present the PDF copy, which shows the intended look.

WHAT'S INSIDE ·

PART	PAGES	USE
Teacher pages	pages 2–12	How to use, overview, choosing a source, pacing, teacher notes, what changed, Logic reference, EQ guide, tutorial links, worked example
Student pages	pages 13–22	Source log, FX notes, Track map, FX log, project sheet (3 pages), rubric, vocabulary, feedback planner
Intro deck (separate file)	deck	18 slides, .pptx + PDF, speaker notes on every slide

NO AUDIO INCLUDED

No audio, video, lyrics or songs are included. Students bring a track of their own or download a multitrack from the Cambridge-MT library themselves (page 4), so you know exactly where every sound came from.

MAKE IT FIT YOUR CLASS ·

- **Source tracks:** students may use their own earlier project, so the project also works without any download.
- **Easier or harder:** my variations on page 6.
- **Feedback board:** my sheets use Padlet for the three questions; any class board works.
- **Grading:** my rubric, /25, the average of the student's and your score (page 20).

IN ONE SENTENCE

Students take a track, EQ and pan every part, add at least five effects with a reason for each, and present their choices.

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FX DEEP DIVE

Project overview

OVERVIEW

DMP2 · Structured

FROM MY DMP2 PROJECT MENU

FX Deep Dive · Structured · Mixing & Effects
Enhance a track using at least 5 audio effects. EQ, pan, and mix every track for a professional result.

WHAT STUDENTS MAKE ·

One Logic Pro mix: an existing track, either a student's own earlier project or a multitrack, with every track EQ'd and panned and at least five different effects chosen for the genre. They hand in a bounced .mp3 or .wav of the final mix, the zipped .logicx project, and a Padlet post with three feedback questions, then walk the class through their FX choices.

WHAT THIS PROJECT REINFORCES ·

This is the primary mixing skills project. It shifts focus entirely to post-production. The EQ Cheatsheet PDF and Plug-in Links doc are key companion resources.

STANDARDS ·

Supports the National Core Arts Standards for Music (Music Technology, high school): refining original work (MU:Cr3.1), presenting it (MU:Pr6.1) and responding to it (MU:Re7.2).

GOALS ·

- Learn and apply at least 5 different audio effects in a real mix
- Develop EQ and panning skills across all tracks in a session
- Understand how FX choices relate to genre and style
- Build a polished, balanced mix with clarity and depth

AT A GLANCE ·

Grades	9–12: the second-year course, after a first year of Logic Pro basics
Length	7 class periods: review and choose, four working days, bounce, present
Software	Logic Pro: Channel EQ, Compressor, ChromaVerb or Space Designer, Echo, Chorus and other plug-ins, pan, sends, bounce
You supply	Source tracks (students' own projects, or multitracks they download), headphones, a feedback board, and speakers if you can
Assessment	My rubric, 5 categories scored 1–5, /25; final grade = average of the student's and the teacher's score (page 20)

BEFORE THEY START

Students should already know how to open a project, work in the Mixer, and save, bounce and zip in Logic. My project uses the 11 FX they took notes on in class (listed on page 14); the Logic quick reference (page 9) covers every tool this project uses.

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FX DEEP DIVE

Setup and choosing a source track

SETUP

Teacher

YOU NEED ·

- **A Mac with Logic Pro** for each student (a current version), and **headphones**. Speakers help for the headphones-and-speakers check in step 5.
- **A source track for every student**: their own earlier project, or a multitrack they download. Nothing is included.
- **A place to submit** an .mp3 or .wav and a zipped .logicx project (your class page).
- **A feedback board** for the three questions (my sheets use Padlet).
- **Printed student pages**, one set each (pages 13–22).

BEFORE CLASS 1 ·

- [ ] Pre-select 5–10 multitrack sessions with sensible track counts (8–16), as my notes suggest, or decide to let students choose.
- [ ] Read the Cambridge-MT library's terms of use so you can tell students what they may do with a download.
- [ ] Mix a short stretch of your own track twice, dry and with effects, for slide 2.
- [ ] Print the student pages; set up the submission spot and the feedback board.

CHOOSING A SOURCE ·

My project gives two choices. Page 13 is the student log for either one.

- **Remix your own project.** No download is needed, and the student already knows the track. Check that it has at least five separate tracks, so there is something to EQ and pan.
- **A multitrack from Cambridge-MT.** The library's sessions are separate audio files for each instrument. Students download one from the library's own site and import it into Logic (page 9). This kit does not include or redistribute any of them.

PRE-SELECTING SESSIONS

- Aim for 8–16 tracks; a 40-track session is a different project.
- Download each one yourself and open it in Logic before students do.
- Note the genre of each, so students can pick one whose style they can describe.
- Keep a short list on the board, with the track count beside each.

TERMS AND CREDIT

Libraries set their own rules for what you may do with their files, such as using them to practice mixing but not sharing them. Check before students post a mix publicly, and credit the source if the library asks.

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FX DEEP DIVE

Pacing guide · 7 class periods

PACING

Teacher

7 class periods, as on my project menu. The “My plan” column is my day-by-day pacing guide as I wrote it; the other columns show what to put on the screen and in students’ hands.

CLASS	MY PLAN	TEACHER MOVES	DECK	STUDENT PAGES
CLASS 1	**Class 1:** Review the 11 FX from class notes. Listen to examples. Students choose their source.	Slides 1–6: hook (dry, then mixed), what effects do, the seven classes, steps 1 and 2, why each effect needs a reason. Hand out the project sheet, the FX notes and the Source log.	Slides 1–6	p. 13, p. 14, p. 17, p. 18, p. 19, p. 21
CLASS 2	**Classes 2–5:** Core working days — apply FX, EQ all tracks, set panning.	Slides 7–11: the live EQ demo on a soloed track, pan the band, reverb and delay on a send, compression by the numbers. Then work time; circulate.	Slides 7–11	p. 15, p. 16
CLASS 3		Slide 12: EQ and pan every track. Then work time.	Slide 12	—
CLASS 4		Slide 13: add effects with a reason and A/B each one. Then work time.	Slide 13	—
CLASS 5		Full working day: refine the mix. Ask each student to bypass one effect and tell you what changed.	No new slides	—
CLASS 6	**Class 6:** Final adjustments, bounce. Prepare presentations and write 3 Padlet feedback questions.	Slides 14–16: refine and bounce, what you turn in, the rubric. Students self-score, bounce and post their three questions.	Slides 14–16	p. 20, p. 22
CLASS 7	**Class 7:** Presentations with Padlet Q&A peer feedback.	Slides 17–18: presentation day and the close. Every student presents and answers 2 of each classmate’s questions.	Slides 17–18	p. 22

THE ONE RULE FOR CLASS 2

From my common struggles: do the EQ demo early, with dramatic cuts and boosts on a soloed track. Slide 8 is that demo, and the EQ guide (page 10) is your script.

CLASSES 3–5

Working days. Slides 12 and 13 open Classes 3 and 4; Class 5 has no new slide. Circulate and ask: why is that effect there, and what happens when you bypass it? For my mix critique day, use the end of Class 5 or Class 6 to play two or three volunteer mixes on the main speakers.

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FX DEEP DIVE

Teacher notes · the big idea

WHAT THIS PROJECT REINFORCES ·

This is the primary mixing skills project. It shifts focus entirely to post-production. The EQ Cheatsheet PDF and Plug-in Links doc are key companion resources.

COMMON STUDENT STRUGGLES ·

- Students apply FX randomly without purpose — push them to articulate WHY they're using each effect.
- EQ feels invisible to beginners. Do a live demo early with dramatic cuts and boosts on a soloed track.
- Cambridge-MT tracks can be overwhelming. Pre-select 5–10 sessions with reasonable track counts.
- Students skip panning and leave everything center. Play a professional mix and point out the width.
- Some students treat this as a composition project. Redirect early.

SUGGESTED VARIATIONS ·

Easier: Reduce the FX requirement to 3 and provide a pre-mixed reference with notes on which effects were used.

More Advanced: Require 7+ FX including at least one send/bus effect. Have them write a paragraph justifying each FX choice.

WHERE THIS KIT HELPS WITH EACH STRUGGLE

- FX without purpose: the FX log (page 16) asks why, and slide 6 gives students four questions.
- EQ feels invisible: the EQ guide (page 10) is a demo script, and slide 8 is the demo.
- Overwhelming sessions: page 4 helps you pre-select, and the Source log (page 13) has students count tracks.
- Everything panned center: slide 9 and the Track map (page 15).
- Treated as composition: slide 6 and the goals on page 17 keep the focus on FX.

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SUGGESTED RESOURCES ·

MY LIST	IN THIS KIT
10 FX Every Music Producer Needs to Know — Native Instruments (YouTube)	Not included: it is a video on YouTube, so search it by title (link on page 11). Its title says 10; my class covers 11 (page 14 lists them). Page 9 shows where each kind of effect lives in Logic.
Cambridge-MT Multitrack Library	Not included and not mine to hand out: students download a session from the library's own site. Page 13 helps you pre-select sessions and reminds students to read the library's terms.
EQ Cheatsheet PDF (in course materials)	Not included: its cover credits it to Jay Dwivedi, so it is not mine to hand out. Use your own copy beside page 10, this kit's own EQ guide and live-demo plan.
Plug-in Links doc with tutorials for each effect category	Included, from the teacher's own materials: page 11 lists my tutorial links (titles and web addresses only) for each of the 11 FX. Page 9 lists the Logic plug-in for each.

HOW TO GRADE ·

- My rubric: five categories, each scored 1–5, /25 (page 20). Students score themselves first; the final grade is the average of their score and yours.

Suggestions from this kit, not part of my rubric:

- Technical Skill: ask the student to open Channel EQ on one track and say which frequency they moved, whether they boosted or cut it, and why.
- Sound Design & Mix Quality: have the student bypass one effect live and describe what changed (an A/B test). Listen on headphones and on the main speakers.
- Composition & Structure: the source track brings its own form. Look at whether the mix supports it (does the chorus open up?), not whether the student wrote it.
- Documentation & Presentation: the FX log (page 16) and the three Padlet questions are the evidence. My rubric mentions reflection, so you may want to ask for a sentence or two.

NOTES / OBSERVATIONS ·

Great project to revisit the EQ Cheatsheet. Consider a mix critique day where you workshop student projects on the main speakers.

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FX DEEP DIVE

What changed from the original

CHANGES

Teacher

This kit is built from three of my documents: the **Project Sheet & Rubric**, the **Teacher Notes** and the **FX Deep Dive** entry on my DMP2 project menu. This page lists every change to my wording, so you know exactly what is mine and what was added.

WORDING CHANGES ·

- My LMS name is written as “your class page” (checklist items 1 and 2).
- Page numbers are added to steps 1 to 4 (the FX notes, the Source log, the FX log and the Track map).
- My four resources are named as I wrote them (page 7). My Plug-in Links doc is reproduced in this kit’s layout as page 11 (titles and web addresses only). The EQ Cheatsheet is by another author, so it is named only. The video and the library are named by title only.
- The 11 FX my sheets mention are listed on page 14, from my own FX notes table (the Native Instruments video title says 10; my class covers 11).
- My pacing guide’s “Classes 2–5” row spans the Class 2, 3, 4 and 5 rows, so each class can name its slides.
- Layout only: the rubric’s Name and Date lines moved to the top; its Notes box became ruled lines.
- The intro deck shortens my wording (full wording on pages 17–19): slide 5 shortens steps 1 and 2, slide 12 shortens step 4, slide 13 shortens steps 3 and 5, and slide 14 shortens steps 5 and 6.

KEPT AS I WROTE IT ·

Everything else from my three documents is kept as I wrote it: the overview, goals, the six vocabulary terms, the five requirements, steps (apart from the page numbers), tips, checklist and file name (“Your Name – FX Deep Dive”), the rubric (five categories scored 1–5, /25, the average of the student’s and the teacher’s score), what the project reinforces, the five common struggles, the variations, my day-by-day pacing and my closing note, and the project menu entry quoted on page 3. No requirement or rubric row is added.

NEW IN THIS KIT ·

Everything else is new and supports my sheets without changing them: how to use the kit, the overview and setup pages (including how to choose a source track), the teacher moves and slide numbers in the pacing guide, the grading suggestions, the Logic quick reference, the EQ guide and demo plan, the worked example (an invented eight-track session described on paper, not a recording), the Source log, the FX notes page, the Track map, the FX log, the vocabulary page, the presentation planner and the intro deck. None of these adds a requirement or a rubric row. The tutorial links page and the list of the 11 FX come from my own materials, laid out for this kit.

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FX DEEP DIVE

Logic quick reference

SETUP

Teacher + student

Everything students need in Logic for this project, in one table. Demonstrate EQ (slide 8) and sends (slide 10), then let students work. Names are from current Logic Pro versions.

Import a multitrack	File › Import › Audio File (Shift-Command-I), or drag the audio files onto the Tracks area. Set the project tempo if the session came with one.
Mixer	Press X to show or hide the Mixer. Each track has a channel strip with an EQ display, Audio FX slots, Sends, a Pan knob and a fader.
Add an effect	Click an empty Audio FX slot on a channel strip and choose a plug-in from the menu. Plug-ins are grouped by kind: EQ, Dynamics, Reverb, Delay, Distortion, Modulation.
EQ	Double-click the EQ display on the channel strip to open Channel EQ. Drag a band up to boost or down to cut; the Q setting sets how wide the move is.
Compression	Audio FX slot › Dynamics › Compressor. Lower the Threshold until the meter shows gain reduction, then set the Ratio.
Reverb and delay	Reverb › ChromaVerb (or Space Designer). Delay › Echo or Stereo Delay. Try a delay time that matches the tempo, such as 1/8 note.
Chorus, phaser, flanger	Modulation › Chorus, Phaser or Flanger. These move the sound over time; use a small amount.
Distortion	Distortion › Distortion II or Overdrive. A little adds warmth; a lot is an effect on its own.
Send and bus	Click a Send slot and choose Bus 1; an Aux channel appears. Put a reverb or delay on the Aux, then turn each track's send knob up to taste.
Pan	Drag the Pan knob under the fader; Option-click it to return to center.
A/B test	Click the plug-in's Bypass button in its window title bar to hear the track without it, then switch it back on. Use Solo (S) to hear one track alone.
Bounce and zip	File › Bounce › Project or Section (Command-B), MP3 or WAV. In Finder, Control-click the .logicx and choose Compress.

VERSION NOTE

Menu names are from current Logic Pro versions; older versions may differ slightly. Logic Pro is a trademark of Apple Inc., named only to describe what the project needs. This resource is not affiliated with Apple.

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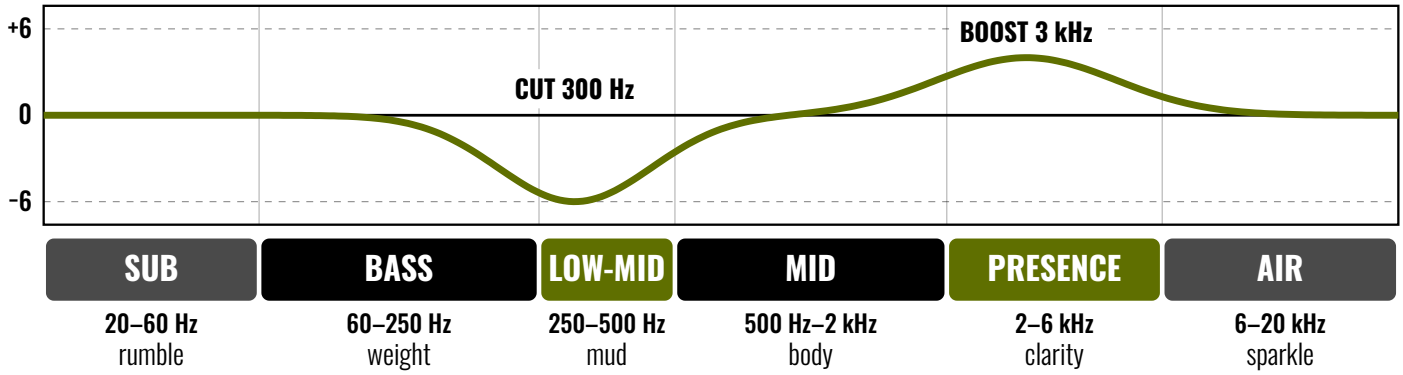
FX DEEP DIVE

EQ guide and live demo plan

EQ GUIDE

Teacher + student

My notes say EQ feels invisible to beginners, so do a live demo early. This page is a one-page guide you can project or print, and the plan for a five-minute demo (slide 8). The bands and fixes are rules of thumb, not laws: tell students to trust their ears.



COMMON PROBLEMS AND A FIRST TRY

IT SOUNDS…	TRY
Muddy or boomy	Cut a little around 200–400 Hz
Boxy, closed in	Cut a little around 400–800 Hz
Harsh or sharp	Cut a little around 2–5 kHz
Thin or weak	Boost a little around 100–250 Hz
Dull, no sparkle	Boost the highs, 6 kHz and up
Rumble you cannot hear	High-pass below 30–80 Hz

THE FIVE-MINUTE DEMO

1. Pick a drum or bass track from your own project and **solo** it.
2. Open Channel EQ. Make a narrow band and **boost 12 dB**. Sweep it slowly from low to high.
3. Stop on a spot that sounds bad. **Cut** it by 6 dB.
4. Bypass the EQ and back: that is A/B.
5. Un-solo. Ask: did the mix get clearer?

FOUR KINDS OF EQ BAND

BAND	WHAT IT DOES
Bell	Boosts or cuts around one frequency. The Q setting makes it narrow or wide.
Shelf	Boosts or cuts everything above (high shelf) or below (low shelf) a frequency.
High-pass	Removes everything below a frequency. Good for clearing rumble from tracks that do not need low end.
Low-pass	Removes everything above a frequency. Good for making a sound darker.

EQ RULES FOR STUDENTS

Cut before you boost. Make small moves. Listen to the whole mix, not only the solo.

PANNING

Keep the kick, snare, bass and lead vocal near the center. Spread other parts and balance the two sides (page 12).

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FX DEEP DIVE

Tutorial links for each effect

LINKS

Teacher + student

My own list of tutorials, one group for each of the 11 FX (page 14), from the teacher's own materials. Each is a web address to type in, or search the title. Nothing is embedded or included; the sites belong to their makers and addresses can change.

PANNING

Panning for a Wider Mix

youtube.com/watch?v=xT1tQgqBWyU

DELAY

Sound Basics - Reverb & Delay

youtube.com/watch?v=-jPPJEHMepA

How to use a Delay Effect

youtube.com/watch?v=boCZejIYUJg

EQ

Sound Basics - EQ

youtube.com/watch?v=I6ZF_NHvqzU

EQ Explained in 4 minutes

youtube.com/watch?v=34lHk6lfpkI

How to Use EQ: 10 Amazing Tips for a Professional Mix

musicianonamission.com/approach-equalization-two-types-eq/

The Ultimate EQ Cheat Sheet for Every Common Instrument

blog.sonicbids.com/the-ultimate-eq-cheat-sheet-for-every-common-instrument

How to hear EQ

youtu.be/U40thp61nrE

COMPRESSION

Compressors Explained

youtube.com/watch?v=IbIC7B4BU6g

Beginner's Guide to Compression, Part 1

youtube.com/watch?v=BIVfpsoPnOo

Part 2

youtube.com/watch?v=ebHGH3k7g4I

Part 3

youtube.com/watch?v=FE_8-Q6CZXE

DISTORTION

Distortion As An Active Production Element

youtube.com/watch?v=KCcDBHKfp5g

Using Distortion Plug-ins

youtube.com/watch?v=c9NbXzdnVB8

REVERB

Reverb and Delay Explained – Sound Basics

youtube.com/watch?v=-jPPJEHMepA

How to use Reverb

youtube.com/watch?v=JxG3PQhCH0M

What is Reverb? The Beginner's Guide to Reverb in Music Production

blog.landr.com/what-is-reverb/

CHORUS

Chorus Effect Tutorial: How To Understand & Apply A Chorus Effect For Fuller Vocals

youtube.com/watch?v=l0BtzSXw538

PHASER

How To Use The Phaser Tool

youtube.com/watch?v=-lpL9pv_4Eo

PITCH CORRECTION

Pitch Correction & Mastering

youtube.com/watch?v=E4eUe4IWM0U

TREMOLO

How To Use Tremolo

youtube.com/watch?v=ltwb3-lW-GE

FILTER

Exploring the AutoFilter

youtube.com/watch?v=CvnZTAnLyTY

ALSO ON MY LIST

10 FX every music producer needs to know | Native Instruments

youtube.com/watch?v=IkiJMTa6lWo

Multi Track Mixing Stems (Cambridge-MT)

cambridge-mt.com/ms/mtk/

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FX DEEP DIVE

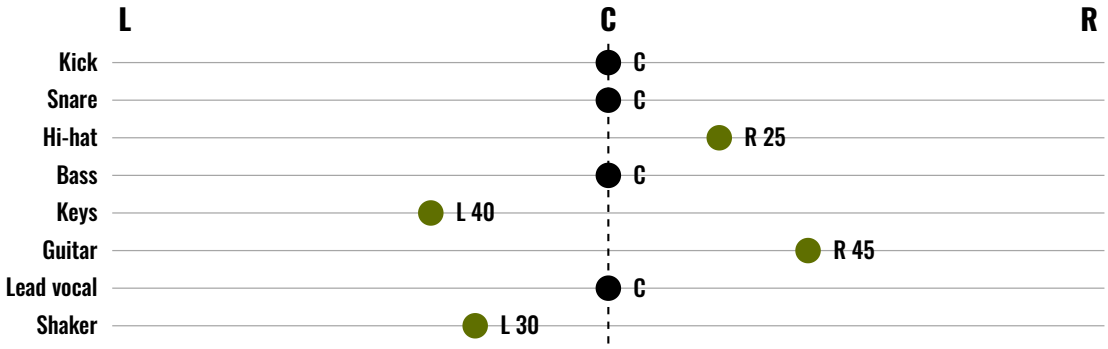
Worked example · an invented session

EXAMPLE

Teacher

An **invented eight-track session, described on paper**: no audio is included, so the numbers are easy to check. Students' sessions will differ.

1 · PLACE THE BAND ·



Left tracks: keys 40 + shaker 30 = 70. Right tracks: hi-hat 25 + guitar 45 = 70. The sides balance.

2 · THE PLAN FOR EACH TRACK ·

TRACK	PAN	EQ	EFFECTS
Kick	center	Boost 60 Hz +3 dB; cut 300 Hz −4 dB	Compressor
Snare	center	Cut 400 Hz −3 dB; boost 4 kHz +3 dB	Reverb (short, on a send)
Hi-hat	right 25	High-pass at 300 Hz	None
Bass	center	Cut 250 Hz −3 dB; boost 80 Hz +2 dB	Compressor
Keys	left 40	High-pass at 150 Hz; cut 300 Hz −3 dB	Chorus
Guitar	right 45	Cut 250 Hz −3 dB; boost 3 kHz +2 dB	Delay (1/8 note)
Lead vocal	center	High-pass at 100 Hz; boost 3 kHz +3 dB	Compressor, reverb, delay (dotted 1/8)
Shaker	left 30	High-pass at 500 Hz	None

Five different effects are used: EQ, compressor, reverb, delay and chorus.

3 · DELAY TIMES AT 100 BPM

NOTE	DELAY TIME
Quarter note	**600 ms**
Eighth note	**300 ms**
Dotted eighth	**450 ms**
Sixteenth note	**150 ms**

One beat = 60 000 ÷ 100 = 600 ms.

4 · COMPRESSION, BY THE NUMBERS

Threshold −18 dB, ratio 4:1. A peak at −6 dB is 12 dB over the threshold. 12 ÷ 4 = 3 dB over, so it comes out at −18 + 3 = −15 dB. Gain reduction = 9 dB (slide 11).

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FX DEEP DIVE

STUDENT

Source log

Class 1

Name _____ Date _____

My source: ☐ **my own previous project** ☐ **a Cambridge-MT multitrack**

Name of the project or song: _____

Genre: _____ **Tempo:** _____ **Tracks:** _____

Count the tracks. Aim for 8–16. If there are many more, ask your teacher before you start.

#	TRACK NAME	KIND OF SOUND	WHAT I THINK IT NEEDS
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			

Genre conventions I should respect (reverb? width? how loud?): _____

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FX DEEP DIVE

FX notes

STUDENT

Class 1

Name _____ Date _____

These are the 11 FX from the class notes. For each one, write in your own words what it does to the sound and where you might use it. Then circle the ones you plan to use: you need at least 5 different effects. Tutorial links for each are on page 11.

#	EFFECT	WHAT IT DOES TO THE SOUND	WHERE I MIGHT USE IT
1	**Panning**		
2	**Delay**		
3	**EQ**		
4	**Compression**		
5	**Distortion**		
6	**Reverb**		
7	**Chorus**		
8	**Phaser**		
9	**Pitch Correction**		
10	**Tremolo**		
11	**Filter**		

RULE

Use at least 5 different FX from the 11 covered in your class notes.

CHECK

Can you say, for each circled effect, what it does and why it fits your genre?

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FX DEEP DIVE

Track map

STUDENT

Classes 2–5

Name _____ Date _____

One row per track. Every track in your mix must be EQ'd and panned thoughtfully. Tick the boxes when it is done, and write down what you did and why.

TRACK	EQ'D	EQ: WHERE, BOOST OR CUT, WHY	PANNED	PAN (L / C / R)	EFFECTS ON IT
	☐		☐		
	☐		☐		
	☐		☐		
	☐		☐		
	☐		☐		
	☐		☐		
	☐		☐		
	☐		☐		
	☐		☐		
	☐		☐		
	☐		☐		
	☐		☐		

MIX CHECK (STEP 5) ·

- ☐ Listened on headphones
- ☐ Listened on speakers, or on a different pair of headphones
- ☐ Left and right feel balanced
- ☐ I can hear every important part
- ☐ I bypassed each effect once to check it helps

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FX DEEP DIVE

FX log

STUDENT

Classes 2–7

Name _____ Date _____

A planning page for step 3, and your notes for the presentation. For each effect, write the reason before you move on. If you break the usual rules of your genre, write why.

FX 1

Effect and plug-in: _____ **On track:** _____

The reason (what problem or feeling it solves): _____

Main settings: _____

Bypass test: what changed? _____

FX 2

Effect and plug-in: _____ **On track:** _____

The reason (what problem or feeling it solves): _____

Main settings: _____

Bypass test: what changed? _____

FX 3

Effect and plug-in: _____ **On track:** _____

The reason (what problem or feeling it solves): _____

Main settings: _____

Bypass test: what changed? _____

FX 4

Effect and plug-in: _____ **On track:** _____

The reason (what problem or feeling it solves): _____

Main settings: _____

Bypass test: what changed? _____

FX 5

Effect and plug-in: _____ **On track:** _____

The reason (what problem or feeling it solves): _____

Main settings: _____

Bypass test: what changed? _____

My favorite change, and why: _____

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Name _____ Date _____

FX DEEP DIVE

Digital Music Production | *Structured Project* | 7 class periods

OVERVIEW ·

Time to level up your mixing skills. Take an existing track — either one of your own previous projects or a song from the Cambridge-MT Multitrack Library — and enhance it using at least 5 different audio effects. Your goal is to use FX creatively and intentionally to shape tone, depth, and space in a professional-sounding mix.

GOALS ·

- Learn and apply at least 5 different audio effects in a real mix
- Develop EQ and panning skills across all tracks in a session
- Understand how FX choices relate to genre and style
- Build a polished, balanced mix with clarity and depth

KEY VOCABULARY ·

EQ (Equalization)	A tool that adjusts the volume of specific frequencies to shape tone and clarity
Compression	Reduces the dynamic range — making quiet parts louder and loud parts quieter
Reverb	Simulates the natural reflections of a space, adding depth and ambience
Delay	Repeats a sound after a set time, creating echo or rhythmic effects
Panning	Placing sounds left or right in the stereo field to create width and separation
Modulation	Effects like chorus, phaser, and flanger that add movement by altering a sound over time

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FX DEEP DIVE

Project sheet · requirements and steps

STUDENT

Name _____ Date _____

REQUIREMENTS ·

- ☐ Choose your source: remix your own previous project OR pick a multitrack from Cambridge-MT
- ☐ Use at least 5 different FX from the 11 covered in your class notes
- ☐ Every track in your mix must be EQ'd and panned thoughtfully
- ☐ FX choices should be stylistically appropriate for the genre
- ☐ Your final mix should demonstrate clarity, depth, and balance

STEPS / PROCESS ·

1. **Review** — Revisit your notes on the 11 FX from the Native Instruments video (notes page: page 14)
2. **Choose your source** — Decide whether to remix your own project or download a multitrack from Cambridge-MT (log it on page 13)
3. **Plan and apply FX** — Add at least 5 different effects across your mix. Think about genre conventions. (FX log: page 16)
4. **EQ and pan every track** — Shape tone with EQ and place each instrument in the stereo field (Track map: page 15)
5. **Refine your mix** — Adjust levels, tweak FX settings, and check on headphones and speakers
6. **Finalize** — Bounce your final mix and prepare to walk the class through your FX choices
7. **Present & Get Feedback** — Create your own Padlet post with 3 questions you want classmates to answer about your project. Present to the class. During classmates' presentations, answer at least 2 of their 3 questions on their Padlet post.

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FX DEEP DIVE

Project sheet · tips and checklist

STUDENT

Name _____ Date _____

TIPS FOR SUCCESS ·

- Less is more. A well-placed reverb or delay can transform a track. Piling on effects usually makes it worse.
- Always A/B test your effects — bypass the plug-in and listen with and without it.
- EQ is your most powerful tool. Get your EQ right before adding other effects.
- If using Cambridge-MT, pick a song with 8–16 tracks. Don't pick a 40-track session.

SUBMISSION CHECKLIST ·

- ☐ Bounced .mp3 or .wav of final mix uploaded to your class page
- ☐ Zipped .logicx project file uploaded to your class page
- ☐ Files named correctly: "Your Name – FX Deep Dive"
- ☐ At least 5 different FX used and ready to discuss
- ☐ All tracks EQ'd and panned
- ☐ Padlet post created with your 3 feedback questions
- ☐ Answered at least 2 questions on classmates' Padlet posts during presentations

NOTES ·

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FX DEEP DIVE

Rubric

RUBRIC

/ 25

Name _____ Date _____

GRADING RUBRIC /25

FX Deep Dive | Structured Project

Each category is scored 1–5. Your final grade is the average of your self-score and your teacher's score.

Category	1	2	3	4	5
CREATIVITY & ORIGINALITY	Minimal creative effort; relies on defaults or clichés	Some creative ideas but largely predictable	Shows originality; experiments with sound or style	Highly creative; unique approach with clear artistic intent	Exceptional creativity; innovative, bold, and expressive
TECHNICAL SKILL / DAW USE	Limited understanding of Logic Pro tools	Basic tool use with frequent errors	Competent workflow; uses tools correctly	Advanced use of Logic features and shortcuts	Expert-level command of multiple Logic systems
COMPOSITION & STRUCTURE	Disorganized; no clear form or direction	Some structure present but underdeveloped	Clear, cohesive form with logical sections	Engaging structure with dynamic progression	Professional pacing, flow, and musical arc
SOUND DESIGN & MIX QUALITY	Poor sound choices; unbalanced or muddy mix	Some variety but inconsistent levels and EQ	Good sound selection; mostly balanced mix	Strong sound palette; clean, well-balanced mix	Exceptional sound design; professional-quality mix
DOCUMENTATION & PRESENTATION	Missing or incomplete; no reflection	Minimal documentation with little insight	Satisfactory; provides basic detail and reflection	Well-documented; clear and thoughtful reflection	Exceptional; thorough insight and polished delivery

Student Score: _____ / 25	Teacher Score: _____ / 25
Final Grade = Average of Student + Teacher = _____ / 25	

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FX DEEP DIVE

Key vocabulary

STUDENT

Vocabulary

Name _____ Date _____

The six words from your project sheet, with my definitions. Rewrite each in your own words, then show where it happens in your own project.

EQ (EQUALIZATION) — A tool that adjusts the volume of specific frequencies to shape tone and clarity

In my own words: _____

Where it shows up in my mix: _____

COMPRESSION — Reduces the dynamic range — making quiet parts louder and loud parts quieter

In my own words: _____

Where it shows up in my mix: _____

REVERB — Simulates the natural reflections of a space, adding depth and ambience

In my own words: _____

Where it shows up in my mix: _____

DELAY — Repeats a sound after a set time, creating echo or rhythmic effects

In my own words: _____

Where it shows up in my mix: _____

PANNING — Placing sounds left or right in the stereo field to create width and separation

In my own words: _____

Where it shows up in my mix: _____

MODULATION — Effects like chorus, phaser, and flanger that add movement by altering a sound over time

In my own words: _____

Where it shows up in my mix: _____

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FX DEEP DIVE

Present and get feedback

STUDENT

Classes 6–7

Name _____ Date _____

FROM YOUR PROJECT SHEET, STEP 7

Present & Get Feedback — Create your own Padlet post with 3 questions you want classmates to answer about your project. Present to the class. During classmates' presentations, answer at least 2 of their 3 questions on their Padlet post.

MY THREE FEEDBACK QUESTIONS ·

Ask about things you really want to know, not yes-or-no questions.

Question 1: _____

Question 2: _____

Question 3: _____

CLASSMATES' QUESTIONS I ANSWERED (AT LEAST 2) ·

CLASSMATE	THEIR QUESTION	MY ANSWER

BEFORE I PRESENT ·

- ☐ Files named "Your Name – FX Deep Dive"; .mp3 or .wav and zipped .logicx uploaded
- ☐ Padlet post created with my 3 questions
- ☐ Rubric self-score done (page 20)

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QUESTIONS?

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CREDITS

- Project sheet, rubric, teacher notes and intro deck: original work by Brian Lotze, Hudson Beat.
- Worked example (an invented eight-track session described on paper), the EQ map, pan map and compressor drawings and layouts: original work, Hudson Beat.
- No audio, video, lyrics, album art, chart data or multitrack sessions are included. Cambridge-MT and Native Instruments are named only as the owner's resource list names them; they are not affiliated with or endorsing this resource. The list of the 11 FX and the tutorial links page are from the teacher's own materials; the linked sites and videos belong to their makers and are not included. The EQ Cheatsheet in the teacher's course materials is by another author and is not included. Padlet and YouTube are named only as examples of tools you may already use.
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